

WINDOWS SOC LAB

Windows Event Log Monitoring & Threat Investigation

SOC L1 / Blue Team Portfolio Project

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Project Objective

Built a Windows-based SOC investigation lab to simulate endpoint activity, collect Windows/Sysmon telemetry, investigate process creation events, identify suspicious activity, and document findings using a SOC analyst workflow.

1. Project Overview

- This project demonstrates a basic Windows endpoint monitoring and threat investigation workflow from a SOC L1 perspective.
- A Windows 11 virtual machine was configured as a controlled SOC laboratory environment. Sysmon was deployed to collect detailed process creation telemetry, which was then reviewed through Windows Event Viewer.
- The lab generated both normal Windows activity and controlled suspicious PowerShell activity. The resulting events were collected as evidence and investigated to determine whether the activity was benign or suspicious.
- The project demonstrates the following SOC workflow:
- Generate Activity → Collect Logs → Detect → Investigate → Classify → Document

2. Lab Environment

Component	Details
Host	Personal Windows System
Virtual Machine	Windows 11
VM Name	WIN11-SOC-LAB
Virtualization	Oracle VirtualBox
Monitoring Tool	Microsoft Sysmon
Log Analysis	Windows Event Viewer
Primary Log	Microsoft-Windows-Sysmon\Operational
Main Event Investigated	Event ID 1 – Process Creation

3. Tools Used

- Oracle VirtualBox
Used to create and run the isolated Windows 11 SOC laboratory environment.
- Windows 11
Used as the monitored endpoint.
- Microsoft Sysmon
Used to collect detailed endpoint telemetry, including process creation, command lines, users, hashes, and parent-child process relationships.
- Windows Event Viewer
Used to review and export Sysmon events for investigation.

4. Detection & Evidence Collection

4.1 Sysmon Event Monitoring

Sysmon was configured on the Windows 11 laboratory endpoint to collect process creation telemetry. **Event ID 1 — Process Creation** was used as the primary event for monitoring and investigation.

The collected telemetry included process names, executable paths, command lines, users, integrity levels, hashes, and parent-child process relationships.

4.2 Activity Generation

The lab generated both legitimate Windows activity and controlled suspicious-looking activity for investigation purposes.

Three process creation events were selected for analysis:

- **Evidence 01 — svchost.exe:** Legitimate Windows service activity.
- **Evidence 02 — WmiPrvSE.exe:** Legitimate Windows WMI Provider Host activity.
- **Evidence 03 — powershell.exe:** Controlled PowerShell execution generated specifically for the SOC lab.

4.3 Evidence Collection

The relevant Sysmon Event ID 1 events were reviewed using Windows Event Viewer. The selected events were saved as .evtx evidence files and supporting screenshots were captured for documentation.

The evidence was organized in the project **03-Evidence** folder.

Evidence Summary

Evidence	Process	Event	Initial Classification
Evidence 01	Svchost.exe	Event ID 1	Benign
Evidence 02	WmiPrvSE.exe	Event ID 1	Benign
Evidence 03	Powershell.exe	Event ID 1	Suspicious/Simulated SOC Activity

5. Investigation Findings

5.1 Investigation 01 — SVCHOST.EXE

- **Finding:**

The Sysmon Event ID 1 event showed svchost.exe executing from the legitimate C:\Windows\System32\ directory under the NT AUTHORITY\SYSTEM account.

- **Analysis:**

The command line indicated that the process was hosting the legitimate wldsvs Windows service. The executable path, Microsoft publisher information, and execution context were consistent with normal Windows activity. No obvious suspicious indicators were identified.

- **Classification:**

Benign

- **SOC Conclusion:**

No malicious activity was identified from this event alone.

The activity was assessed as legitimate Windows service execution.

5.2 Investigation 02 — WMIPRVSE.EXE

- **Finding:** Sysmon Event ID 1 recorded WmiPrvSE.exe executing from the legitimate C:\Windows\System32\wbem\ directory under the NT AUTHORITY\NETWORK SERVICE account.
- **Analysis:** The process used the normal -secured - Embedding parameters associated with Windows WMI Provider Host activity. The executable path, Microsoft publisher information, and execution context were consistent with legitimate Windows activity. No obvious suspicious indicators were identified.
- **Classification:** Benign
- **SOC Conclusion:** No malicious activity was identified from this event alone. The activity was assessed as legitimate Windows WMI Provider Host execution.

5.3 Investigation 03 — POWERSHELL.EXE

- **Finding:** Sysmon Event ID 1 recorded powershell.exe executing from the legitimate C:\Windows\System32\WindowsPowerShell\v1.0\ directory under the WIN11-SOC-LAB\shali user account.
- **Analysis:** The process was executed with the -NoProfile and -Command parameters to directly run a PowerShell command. Although the executable and path were legitimate, the use of PowerShell with command-line execution was treated as suspicious activity for the purposes of this controlled SOC lab. The command only

used Write-Host and did not perform a malicious action. The activity was intentionally generated for the investigation.

- **Classification:** Suspicious / Simulated SOC Activity
- **SOC Conclusion:** Suspicious PowerShell execution was detected and investigated. No malicious behavior was identified from this event alone. The event demonstrates how Sysmon Event ID 1 can provide useful process execution and parent-child relationship telemetry for SOC investigation.

6. MITRE ATT&CK Mapping

6.1 PowerShell Execution

- **Technique:** T1059.001 — PowerShell
- **Observation:** The Sysmon event recorded execution of powershell.exe with the -NoProfile and -Command parameters.
- **Relevance:** PowerShell is a legitimate Windows administration tool but is also commonly monitored by SOC teams because attackers can abuse it to execute commands on compromised systems.
- **Lab Context:** In this project, the PowerShell activity was intentionally generated as controlled simulated activity. The command executed only Write-Host and did not perform malicious actions.

7. Overall Findings & Conclusion

7.1 Overall Findings

- **Finding:** The investigation demonstrated that Sysmon Event ID 1 can provide detailed process execution telemetry useful for SOC analysis.
- **Benign Activity:** svchost.exe and WmiPrvSE.exe were identified as legitimate Windows processes based on their expected paths, Microsoft ownership, command-line parameters, and execution context.
- **Suspicious Activity:** powershell.exe was identified as suspicious simulated activity because it used command-line PowerShell execution. The activity was intentionally generated for the lab and did not perform a malicious action.
- **SOC Assessment:** The investigation showed the importance of analyzing process paths, command lines, users, integrity levels, hashes, and parent-child process relationships rather than classifying an event based only on the process name.

7.2 Conclusion

- This project demonstrated a basic Windows SOC monitoring and threat investigation workflow using Windows 11, Sysmon, and Event Viewer. Process creation events were collected, preserved as evidence, analyzed, classified, and documented.
- The investigation demonstrated the ability to distinguish legitimate Windows activity from controlled suspicious activity and showed how endpoint telemetry can support SOC L1 investigation and incident triage.

8. Future Improvements

- **Centralized Log Collection:**

Integrate the Windows endpoint with a SIEM such as Microsoft Sentinel to centralize and correlate security events.

- **Automated Detection:**

Create detection rules and alerts for suspicious process execution, PowerShell activity, and abnormal parent-child process relationships.

- **Expanded Sysmon Monitoring:**

Enable additional Sysmon event types such as network connections, file creation, registry activity, and process access to improve endpoint visibility.

- **Threat Intelligence Integration:**

Enrich observed hashes, IP addresses, domains, and other indicators with threat intelligence sources during investigations.

- **Incident Response Simulation:**

Expand the lab with additional controlled scenarios such as persistence, credential access, suspicious scheduled tasks, and lateral movement.